LLOYD INSTITUTE OF ENGINEERING & TECHNOLOGY

Department of Computer Science & Engineering & Allied Branches

B.TECH SECOND YEAR MINI PROJECT · BCC351

SMARTPREP: AN AI-GUIDED CAREER PREPARATION & PREREQUISITE-DRIVEN ROADMAP OPERATING SYSTEM

Mini Project Presentation – II (Final Evaluation) | Session 2026–27

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OVERVIEW

Presentation Agenda

1. Project Recap
2. Project Progress
3. System Architecture
4. Implementation Details
5. Working Demonstration
6. Technical Understanding
7. Results & Testing
8. Conclusion & Future Scope
9. References

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RECAP

Project Recap (from Review-01)

Problem Statement

- **Information Overload & Dispersion:** Engineering students struggle with scattered, unstructured tutorials across YouTube, blogs, and documentation without coherent learning paths.
- **The Prerequisite Blindspot:** Beginners frequently attempt advanced frameworks (Spring Boot, Next.js, DP) without mastering foundational concepts, leading to high dropout rates.
- **Lack of Dynamic Adaptive Pacing:** Generic online courses enforce rigid timelines without context-aware recovery plans when a student misses study days.
- **Missing Diagnostic Verification:** Absence of integrated, instant diagnostic feedback to detect specific knowledge gaps and conceptual off-by-one errors.

Objectives

01 Curate structured syllabi across 6 key engineering roles (Backend, Full Stack, AI/ML, Data Eng, Android, Placement).

02 Implement a dynamic state machine that locks advanced modules until foundation competencies are verified.

03 Embed targeted technical quizzes with sound/confetti feedback to pinpoint and patch conceptual gaps in real time.

04 Build an offline-first rule engine that generates personalized daily focus directives and 3-day recovery pipelines.

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PROJECT PROGRESS

Planned vs Completed

100%

OVERALL COMPLETION

✓ Full-stack React 19 + Vite 8 Single Page Application architecture fully established and validated.

✓ Automated Prerequisite State Machine: Dynamic stage status resolution (Completed, Current, Upcoming, Locked).

✓ Deterministic Local AI Guidance Kernel: Intent recognition for Daily Study Directives, 3-Day Catch-up Plans, and Weak-Area alerts.

✓ Interactive Diagnostic Knowledge Check: Multi-question assessments with Web Audio API sound synthesis and visual feedback.

✓ Real-Time Progress Tracking Engine: Daily study streak tracker, weekly hour logging, and progress percentage computation.

✓ Enhanced UI/UX: Dark & Light theme switching, responsive desktop sidebar, and mobile bottom navigation.

✓ Production Optimization: Configured for zero-config cloud deployment on Vercel and Railway with instant client builds.

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Architecture

System Architecture / Workflow



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IMPLEMENTATION

Implementation Details

1. Component-Driven Modular UI (React 19 + Tailwind CSS)

Clean architectural separation between presentation (views) and domain logic. Atomic components (Navbar, Sidebar, RoadmapStage, StudySession) guarantee high reusability and responsiveness across mobile and desktop devices.

2. Prerequisite State Machine & DAG Traversal

The learning curriculum is structured as a Directed Acyclic Graph (DAG). Stages evaluate dynamically: COMPLETED (in array), CURRENT (first incomplete module), UPCOMING (next in line), and LOCKED (unmet prerequisite barrier).

3. Deterministic Local AI Guidance Kernel (aiEngine.js)

Operates completely offline with 0ms network latency. The Context Builder aggregates active stage, velocity, deficit, and flagged weak areas, dispatching structured study directives and 3-day recovery pipelines.

4. Interactive Diagnostic Quiz & Web Audio Synthesis

Features multi-question conceptual evaluations. Integrates the Web Audio API for synthetic audio cues without external media assets, coupled with canvas-confetti for positive learning reinforcement.

5. Client Persistence & Production Deployment

Synchronizes reactive state with HTML5 LocalStorage for zero-loss offline continuity. Optimized with Vite 8 for sub-second production builds and pre-configured for Vercel / Railway deployment.

WORKING DEMONSTRATION

Working Demonstration



INPUT PROCESSING OUTPUT

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TECHNICAL UNDERSTANDING

Technology Stack & Core Concepts

- **React 19 (Frontend Architecture):** Powers component rendering, custom hooks (useApp), and reactive UI reconciliation via Fiber architecture.
- **Vite 8.3 (Build & Dev Tooling):** Provides lightning-fast Hot Module Replacement (HMR) and optimized Rollup tree-shaking (<700ms build).
- **Tailwind CSS v4 (Design System):** Responsive styling system utilizing modern CSS variables for instant dark/light mode toggling.
- **React Router DOM v7 (Navigation):** Declarative client-side routing with complete SPA history and catch-all redirect handling.
- **Local AI Kernel Engine:** Lightweight, deterministic heuristic evaluator executing context aggregation without cloud latency or costs.

Core Concept / Algorithm

Each career track is modeled as a topological sequence of nodes. Node i is strictly gated until all predecessors Pred(i) are verified in CompletedStages. Concurrently, the AI Kernel aggregates real-time velocity, active knowledge gaps, and target hours to synthesize adaptive 3-day recovery schedules.

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RESULTS

Results & Testing

Test Case	Input Scenario	Expected System Output	Actual Result / Status
TC-01: Track Init	Select 'Backend Developer' goal	Initialize 7 stages; Stage 1 Current, 2-7 Locked; Progress 0%	✓ PASSED (<2ms latency)
TC-02: Prereq Gating	Complete Stage 1 & Stage 2	Stages 1-2 marked Completed, Stage 3 unlocks, Progress = 29%	✓ PASSED (Dynamic unlock verified)
TC-03: Security Lock	Attempt to access Stage 5 directly	Action gated; Lock notification displayed; Stage remains Locked	✓ PASSED (Integrity preserved)
TC-04: AI Recovery	Query: 'I missed 3 days, adjust plan'	AI Kernel calculates deficit and generates 3-day recovery plan	✓ PASSED (Context-aware output)
TC-05: Diagnostic Quiz	Submit quiz on Java Interfaces & Stack	Calculate score, flag weak areas, synthesize Web Audio feedback	✓ PASSED (Zero audio latency)

Key Result: 100% test pass rate across core modules; <15ms UI response latency; sub-second cold build time (681ms); offline persistence verified.

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CONCLUSION

Conclusion, Challenges & Future Scope

Conclusion

- Successfully designed and implemented SmartPrep, an AI-guided career preparation OS for B.Tech CSE students.
- Eliminated the 'prerequisite trap' through automated topological stage gating across 6 engineering tracks.
- Proved that deterministic local AI guidance provides immediate, reliable study pacing without external server dependency.
- Delivered a zero-latency, production-ready SPA with verified state persistence and responsive design.

Challenges & Solutions

- **State Management Without DB:** Resolved by engineering pure stage derivation functions in React Context synced with LocalStorage.
- **AI Guidance Without API Costs:** Resolved by architecting a rule-based AI Kernel with context extraction and intent classification.
- **Sound Assets Overhead:** Resolved by utilizing browser Web Audio API oscillator nodes for lightweight audio synthesis.

Future Scope

- Cloud Backend: Connect Spring Boot / Node.js backend with PostgreSQL for multi-device sync and user authentication.
- LLM API Integration: Connect Google Gemini API for automated code review, personalized hints, and dynamic interview prep.
- Peer Social Pods: Introduce peer study rooms, campus-wide streak leaderboards, and collaborative coding challenges.
- Multi-Platform Deployment: Continuous delivery pipeline deploying to Vercel, Railway, and GitHub Pages.

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NOTE: All references must be cited in IEEE format.

THANK YOU

Questions & suggestions are welcome

SmartPrep: An AI-Guided Career Preparation Operating System

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